

# Global Aviation Disruption Assessment 2026

## Business Requirements

Client brief defining the assessment objectives, required analysis, reporting needs, and acceptance criteria.

Scenario note: This is a simulated ICAO client assignment, not an official ICAO commission or endorsement.

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## 1 Business background

Within this assignment, the International Civil Aviation Organization (ICAO) requests an assessment of aviation disruption associated with the 2026 Iran-US conflict scenario. The assessment must help the requesting team understand where disruption is concentrated, how passengers and route operations are affected, and which financial and geographic exposures need planning attention.

## 2 Problem statement

Aviation disruption can affect passenger journeys, airport activity, airspace access, route distance, operating costs, and airline financial exposure at the same time. The requesting team needs a consistent assessment of these effects across airlines, countries, regions, and directional corridors. The supplied records describe different aspects of the scenario and do not all cover the same airlines or dates. The analytical team must establish what each dataset represents, produce clear operational and financial comparisons within defensible scopes, and communicate the main concentrations and planning priorities through an interactive report and a final business case study.

## 3 Objectives and intended decisions

- Identify the largest reported operational and passenger burdens.
- Locate airports, airspace areas, and corridors that warrant contingency planning attention.
- Compare airline and regional exposure within the supplied financial scenarios.
- Describe event-date patterns without treating timing alone as proof of cause.
- Provide recommendations that explain which finding supports each proposed planning action.

## 4 Stakeholders and information needs

| Stakeholder                 | Information needed                                                              |
|-----------------------------|---------------------------------------------------------------------------------|
| Requesting ICAO team        | A clear assessment of operational, regional, and financial exposure.            |
| Executive reader            | The main findings, their business meaning, and recommended planning priorities. |
| Operational planning reader | Airport, airspace, passenger, and directional route concentrations.             |
| Financial planning reader   | Separate modeled total-loss and daily-estimate comparisons.                     |
| Analytical delivery team    | Defined questions, metric rules, scope, and acceptance criteria.                |

## 5 Business questions

### Detailed operations

Q01 How many detailed cancellation records are reported?

Q02 How many passengers are affected by those cancellations?

Q03 How do cancellations vary over the available dates?

- Q04 Which cancellation reasons are most frequent?
- Q05 How many detailed reroute records are reported?
- Q06 What additional distance do those reroutes create?
- Q07 What additional fuel cost do those reroutes create?
- Q08 What delay burden do those reroutes create?

### **Routes and corridors**

- Q09 Which directional corridors have the most reroutes?
- Q10 Which corridors have the most cancellations and affected passengers?
- Q11 Which corridors carry the greatest additional distance, fuel cost, and delay?

### **Airports and airspace**

- Q12 Which regions have the highest airport-affected flight counts?
- Q13 Which countries have the highest airport-affected flight counts?
- Q14 Which airports have the highest reported disruption exposure?
- Q15 Which airports have the longest reported disruption records?
- Q16 How does airport disruption vary over time?
- Q17 Which countries have the greatest airspace closure duration?
- Q18 How do airport and airspace exposure differ by region when reported separately?

### **Financial scenarios**

- Q19 Which airlines have the highest modeled total loss?
- Q20 Which regions have the highest modeled total loss?
- Q21 How does modeled loss differ across airline types?
- Q22 Which airlines have the highest estimated daily loss and related exposure?
- Q23 How do the summary and estimate airline populations differ?

### **Event context**

- Q24 How do daily operational counts differ between military-event and other dates within shared coverage?

## **6 Scope**

### **6.1 Included work**

The assignment covers examination of seven supplied datasets, documented cleaning, source-specific PostgreSQL analysis, population reconciliation, and a four-page Power BI report. The final report must connect operational, geographic, financial, and event-context findings to business questions and planning recommendations.

## 6.2 Excluded work

The assignment does not include live monitoring, forecasting, route optimization, operational automation, or causal attribution. A deduplicated total across airport and airspace exposures is outside scope because the supplied records do not identify overlapping flights. Cancellation rates and aircraft occupancy require denominators that are not supplied.

## 7 Required data and population rules

| Business source         | Analytical population        | Required use                                                       |
|-------------------------|------------------------------|--------------------------------------------------------------------|
| Modeled airline summary | Undated modeled summary      | Summary loss, cancellations, reroutes, revenue-loss percentage     |
| Airline daily estimate  | Undated modeled estimate     | Daily loss and estimated passengers, fuel, cancellations, reroutes |
| Airport disruptions     | Simulated airport detail     | Airport flights affected, duration, severity, geography            |
| Airspace closures       | Simulated closure detail     | Closure count, hours, and reported flight exposure                 |
| Conflict context        | Simulated event context      | Event timeline, military records, reported severity                |
| Detailed cancellations  | Simulated operational detail | Cancellation records and affected passengers                       |
| Detailed reroutes       | Simulated operational detail | Reroute records, distance, fuel, and delay                         |

Detailed records, financial summaries, and financial estimates shall remain separately labeled. A total from one population shall not be used as the denominator of another population. Airport and airspace measures shall be displayed separately. Date comparisons shall use documented coverage, and undated financial values shall not imply a daily trend.

## 8 Mandatory requirements

### BR-001

The assessment shall assess operational disruption and passenger impact across the supplied records.

Acceptance criterion: Questions Q01 to Q08 have sourced answers with units and coverage.

### BR-002

The assessment shall identify geographic concentrations for airport and airspace planning.

Acceptance criterion: Questions Q12 to Q18 have separately labeled rankings.

**BR-003**

The assessment shall compare airline and regional financial exposure within each financial scenario.

Acceptance criterion: Questions Q19 to Q23 distinguish modeled summary from daily estimate.

**BR-004**

The assessment shall identify corridors that need contingency planning attention.

Acceptance criterion: Questions Q09 to Q11 retain direction and distinguish cancellation and reroute measures.

**AR-001**

The assessment shall calculate counts, sums, rankings, and descriptive date comparisons from the supplied data.

Acceptance criterion: All Q01 to Q24 have a result reference or a clear coverage explanation.

**AR-002**

The assessment shall compare military-event and other dates only within common operational coverage.

Acceptance criterion: Q24 uses the intersection of dated airport, cancellation, and reroute coverage.

**AR-003**

The assessment shall use the same population for the numerator and denominator of every ratio.

Acceptance criterion: Each ratio names its source and returns blank for a zero denominator.

**DR-001**

The assessment shall preserve original source files and all valid records.

Acceptance criterion: Seven raw files remain unchanged; cleaning preserves every source row.

**DR-002**

The assessment shall document grain, population, date coverage, and authority for every dataset.

Acceptance criterion: Seven source definitions appear in the data assessment and technical reference.

**DR-003**

The assessment shall preserve unresolved locations without inventing countries.

Acceptance criterion: Original event locations remain available and missing countries remain nullable.

**DR-004**

The assessment shall keep summary, estimate, operational, airport, and airspace populations separate.

Acceptance criterion: No shared total or cross-population ratio is presented without evidence of compatibility.

### **RR-001**

The assessment shall provide four report pages covering overview, finance, event context, and regional corridors.

Acceptance criterion: Each page has readable titles, units, and appropriate filtering.

### **RR-002**

The assessment shall make date and geography filter meaning clear.

Acceptance criterion: Origin, destination, entity geography, severity, and undated finance are distinguished.

### **RR-003**

The assessment shall explain selections outside available operational coverage.

Acceptance criterion: Earlier dates are not described as confirmed zero disruption; coverage is visible.

### **RR-004**

The assessment shall present findings and feasible planning recommendations in business language.

Acceptance criterion: Every recommendation points to an analytical finding; no unsupported savings or causal claims appear.

### **TR-001**

The assessment shall provide a reproducible static analytical workflow.

Acceptance criterion: A reviewer can follow Python, PostgreSQL, and Power BI setup instructions using local credentials.

### **TR-002**

The assessment shall reconcile clean files, database loads, reporting views, and report totals.

Acceptance criterion: Expected row counts and material control totals have matching definitions across stages.

### **TR-003**

The assessment shall keep credentials and local state outside published project files.

Acceptance criterion: Release files contain no passwords, local caches, private study notes, or historical reports.

## **9 KPI definitions**

A key performance indicator (KPI) is a defined measure used to summarize a business question. The report must state its population and units. All amounts use United States dollars (USD).

| KPI                    | Business calculation       | Population and date rule         |
|------------------------|----------------------------|----------------------------------|
| Detailed cancellations | Count cancellation records | Cancellation detail; record date |

| KPI                              | Business calculation                              | Population and date rule               |
|----------------------------------|---------------------------------------------------|----------------------------------------|
| Cancellation passengers          | Sum passengers affected                           | Cancellation detail; record date       |
| Detailed reroutes                | Count reroute records                             | Reroute detail; record date            |
| Additional distance              | Sum additional kilometres                         | Reroute detail; record date            |
| Additional fuel cost             | Sum extra fuel cost in USD                        | Reroute detail; record date            |
| Reroute delay                    | Sum delay hours                                   | Reroute detail; record date            |
| Airport flights affected         | Sum reported affected flights                     | Airport records; record date           |
| Airport disruption hours         | Sum reported duration                             | Airport records; record date           |
| Airspace closure hours           | Sum complete closure durations                    | Closure records; closure start date    |
| Airspace flights affected        | Sum reported affected flights                     | Closure records; closure start date    |
| Modeled summary loss             | Sum modeled airline loss in USD                   | Airline summary; undated               |
| Mean revenue loss                | Unweighted mean of airline percentages            | Airline summary; undated               |
| Estimated daily loss             | Sum supplied daily estimates in USD               | Airline estimate; undated              |
| Estimated fuel per reroute       | Estimated fuel cost / estimated reroutes          | Airline estimate only; undated         |
| Estimated passengers per reroute | Estimated passengers / estimated reroutes         | Exposure ratio, not occupancy; undated |
| Military-event comparison        | Mean daily operational counts by event-date group | Shared operational coverage only       |

## 10 Assumptions constraints and dependencies

The assignment uses a fixed supplied snapshot. Financial summaries and daily estimates have no reporting dates, and their estimation methods are not provided. Flight numbers are not assumed to identify unique events across datasets. Conflict locations may be country-specific, regional, or unresolved. Initial data assessment must establish the available coverage before any comparison is interpreted.

Delivery depends on access to the supplied files, a local PostgreSQL environment, and Power BI Desktop. Findings must stay descriptive where the data does not support stronger conclusions. Any requested metric that needs missing information must be narrowed or explained rather than fabricated.

## 11 Deliverables and success criteria

| Deliverable                 | Acceptance criterion                                                                      |
|-----------------------------|-------------------------------------------------------------------------------------------|
| Preliminary Data Assessment | Documents all seven source populations, initial quality checks, and analytical decisions. |
| Technical Documentation     | Explains the reproducible pipeline, transformations, database, SQL, and semantic model.   |
| Power BI report             | Provides four readable pages with correctly labeled measures and meaningful filters.      |
| Executive Case Study Report | Answers the business questions and supports recommendations with verified results.        |

The requesting team should be able to trace each reported figure to its source population, understand why it matters, and identify the proposed planning action. Raw files must remain unchanged, transformations must be reproducible, and published metrics must not mix incompatible populations.

## Appendix Requirements traceability

| Requirement group | Questions             | Expected output                                               |
|-------------------|-----------------------|---------------------------------------------------------------|
| BR-001            | Q01 to Q08            | Executive overview and operational findings                   |
| BR-004            | Q09 to Q11            | Regional corridor page and route findings                     |
| BR-002            | Q12 to Q18            | Airport, closure, and regional findings                       |
| BR-003            | Q19 to Q23            | Financial scenario page and financial findings                |
| AR-002            | Q24                   | Shared-window SQL comparison and event-context interpretation |
| DR-001 to DR-004  | All questions         | Preliminary assessment and documented source rules            |
| RR-001 to RR-004  | All report outputs    | Four report pages and final case study                        |
| TR-001 to TR-003  | All technical outputs | Technical reference and reproducible release                  |